

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	35.8	12	1.7	0.7	0.3	49.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	35.9	11.7	1.5	0	0.1	50.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	35.2	11.9	1.9	1	0.2	49.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	35.5	12	1.5	0.2	0.1	50.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	34.4	13.3	2.1	0.8	0.3	49.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	34.9	12.4	1.4	0.1	0.1	51.1
U2 - Efficacy additive alpha prior Beta(1, 1)	35.4	12.8	1.5	0.6	0.4	49.3
U3 - Efficacy additive alpha prior Beta(1, 1)	36.8	11	1.7	0.1	0.2	50.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.6	6.2	2.3	1.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.8	6	2.2	1.3	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.5	6.3	2.3	1.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.9	6	2.2	1.2	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.5	6.3	2.3	1.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.9	5.9	2.2	1.2	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	14.6	6.4	2.3	1.3	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	14.8	5.9	2.2	1.2	0.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.3	3.2	0.6	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.2	3	0.6	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.3	3.2	0.6	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.2	3.1	0.6	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.3	3.1	0.6	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.3	3	0.6	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	1.3	3.2	0.6	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.3	3	0.6	0.4	0.1	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	27.6	23.4	35.6	11.7	1.1	0.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	32.4	22.2	34.4	9.5	0.9	0.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	28.3	22.4	35.7	11.6	1.4	0.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	32.7	22.3	33.8	9.6	1	0.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.5	23	36.6	11.8	1.4	0.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	34.1	21.8	32.7	9.6	1.2	0.6
U2 - Efficacy additive alpha prior Beta(1, 1)	27.5	23.4	35.2	11.6	1.7	0.6
U3 - Efficacy additive alpha prior Beta(1, 1)	31.7	22.9	34.5	9.3	1	0.6
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.4	8	8.4	3.4	0.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10	8.1	8.3	3.1	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.4	7.9	8.5	3.4	0.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10	8.1	8.2	3.1	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.4	8	8.5	3.3	0.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10	8.1	8.1	3.2	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.4	7.9	8.6	3.3	0.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.9	8.1	8.3	3.1	0.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.4	1.7	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.4	1.7	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.4	1.8	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.4	1.7	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.4	1.7	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.4	1.7	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.4	1.8	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.4	1.7	0.7	0.1	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.2	49	24.8	10	1.5	2.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.2	52.7	22.5	7.1	1.1	2.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.4	47.9	25.9	9.8	1.6	2.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.9	52.7	20.8	8.1	1.1	2.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.1	47.3	27.1	9.7	1.5	2.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.8	52.8	23	6.9	1	2.5
U2 - Efficacy additive alpha prior Beta(1, 1)	12	48.5	25.3	10.1	1.6	2.5
U3 - Efficacy additive alpha prior Beta(1, 1)	13.7	50.7	24.4	7.6	1.1	2.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.1	11.5	6.9	3.4	0.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.3	12.5	6.4	2.9	0.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.1	11.4	7	3.3	0.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.4	12.2	6.5	3	0.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7	11.3	7.2	3.4	0.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.4	12.3	6.5	2.9	0.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	7.1	11.4	6.9	3.4	0.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	7.4	12.3	6.6	2.9	0.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.9	1.7	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.9	1.6	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.9	1.7	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	4	1.6	0.8	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.9	1.8	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.9	1.7	0.8	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.9	1.7	1	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.9	1.7	0.8	0.2	
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.4	7.6	19.6	49	15.2	0.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12	10.4	18.3	46.6	12.5	0.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.6	8.3	19.6	48.8	14.5	0.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.4	10.3	17.8	46.8	12.5	0.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.5	7.2	18.7	49	16.4	0.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.1	9.6	17.4	48.7	12	0.2
U2 - Efficacy additive alpha prior Beta(1, 1)	7.9	8.6	19.7	48.7	14.9	0.2
U3 - Efficacy additive alpha prior Beta(1, 1)	12.4	9.9	18.4	47.6	11.5	0.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.2	4.8	6.8	10.3	3.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.5	4.9	6.9	10.4	3.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.1	4.7	6.9	10.4	3.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.5	4.9	6.8	10.4	3.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.2	4.6	6.9	10.4	3.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.5	4.9	6.8	10.6	3.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.2	4.7	6.9	10.3	3.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.5	4.9	6.9	10.5	3.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3	1.6	2.4	0.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3	1.6	2.3	0.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.9	1.6	2.4	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3	1.6	2.4	0.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.9	1.6	2.4	0.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3	1.6	2.3	0.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.9	1.6	2.4	0.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	3	1.6	2.4	0.7	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.9	33.7	25.8	23.6	10.6	0.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.7	41.2	24.8	18.6	7.3	0.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.6	34.3	26.2	22.8	10.7	0.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.7	38.9	26.3	18.7	7	0.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.8	33.7	27.3	22.1	10.7	0.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.3	39.7	25	19.5	7.1	0.4
U2 - Efficacy additive alpha prior Beta(1, 1)	6.2	32.5	25.8	24	11.1	0.4
U3 - Efficacy additive alpha prior Beta(1, 1)	8.9	40.3	24.2	18.9	7.3	0.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.9	7.1	7.1	7.3	4.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.1	8.1	7.3	7	3.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.9	7.1	7.1	7.3	4.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.1	7.8	7.2	7.1	3.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.9	7.1	7.2	7.2	4.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.1	7.9	7.2	7.1	3.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.9	7.1	7.3	7.3	4.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.1	7.9	7.3	7.1	3.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.2	1.9	2.4	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.2	2	2.2	1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.1	1.9	2.3	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.2	2	2.2	1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.2	1.9	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.2	2	2.2	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.1	1.9	2.3	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.2	2	2.2	1	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	64.2	17	3.4	1.4	0.4	13.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	65.7	16.6	2.6	1.1	0.3	13.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	62.6	17.9	3.8	1.6	0.4	13.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	66	16.5	2.5	1.2	0.2	13.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	62.9	17.8	4	1.1	0.5	13.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	65.9	15.8	2.5	1.3	0.3	14.2
U2 - Efficacy additive alpha prior Beta(1, 1)	61.5	18.6	3.7	1.9	0.4	13.9
U3 - Efficacy additive alpha prior Beta(1, 1)	65.6	17	2.2	0.9	0.3	14
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.8	6.4	2.2	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.2	6.3	1.9	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.7	6.5	2.2	1	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.3	6.4	1.9	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.8	6.3	2.2	1	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.2	6.3	1.8	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	17.7	6.5	2.2	1	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	18.2	6.3	1.9	0.9	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	2.6	0.6	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	2.5	0.5	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	2.6	0.6	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	2.5	0.5	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	2.6	0.6	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	2.5	0.5	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	2.6	0.6	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	2.5	0.5	0.2	0	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.9	47.9	26.1	9	2.3	0.8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.8	54.1	22.1	6.6	0.6	0.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.5	49.6	23.7	9.8	1.5	0.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.5	54.6	22	6	0.8	1.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.2	49.6	24.7	9.4	2.1	1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.3	54.4	21.4	6.5	0.6	0.8
U2 - Efficacy additive alpha prior Beta(1, 1)	13.8	49.2	26.1	8.5	1.6	0.8
U3 - Efficacy additive alpha prior Beta(1, 1)	16.3	54.4	21.7	6.3	0.5	0.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.8	12.6	6.4	2.4	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.2	13.8	5.8	1.8	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.9	13	6.2	2.3	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.2	14	5.7	1.7	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.9	12.8	6.3	2.3	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.2	14	5.6	1.8	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	7.9	12.9	6.2	2.3	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	8.2	13.9	5.7	1.8	0.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.5	1.4	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.6	1.2	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.5	1.4	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.6	1.2	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.5	1.4	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.7	1.2	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.6	1.4	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.6	1.2	0.5	0.1	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	46.8	11.1	2.7	1.1	0.4	37.9
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	48.6	10.6	2.2	0.9	0.2	37.5
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	46.3	11.7	2.3	1.2	0.4	38.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	48.2	10.4	2.4	0.8	0.2	38
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	45.8	11.6	2.9	1.2	0.5	38
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	48.8	10.9	2.1	1	0.2	37
U2 - Efficacy additive alpha prior Beta(1, 1)	46.9	11.2	2.9	1.3	0.4	37.3
U3 - Efficacy additive alpha prior Beta(1, 1)	47.5	11.3	2.5	0.6	0.2	37.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	16	6.1	2	1	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.2	5.9	2	1	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.8	6.1	2.1	1	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.2	5.8	2	1	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.9	6	2.1	1	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.2	5.9	2	1	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	16	6.1	2	0.9	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	16.2	5.9	1.9	1	0.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	2.7	0.5	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	2.6	0.5	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	2.7	0.5	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	2.6	0.5	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	2.7	0.5	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	2.6	0.5	0.3	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	1	2.7	0.5	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	1	2.7	0.5	0.3	0	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.5	17.2	22.1	13.8	23.8	4.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	24	19.4	19.2	12.9	19.7	4.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.4	17.6	22	15.5	22.9	4.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	23.3	20.2	18.9	12.4	20.2	5
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.1	17.3	22.1	13.8	24.2	4.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	25.4	18.3	20.3	11.9	19.2	4.9
U2 - Efficacy additive alpha prior Beta(1, 1)	18.8	16.4	21.1	14.9	24.6	4.2
U3 - Efficacy additive alpha prior Beta(1, 1)	24.5	18.9	18.8	13.1	19.6	5.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.3	6.4	5.8	5.2	4.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8	6.3	5.7	5.2	4.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.3	6.4	6	5.2	4.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8	6.3	5.6	5.1	4.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.4	6.4	5.7	5.2	4.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8	6.3	5.8	5.2	4.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	7.3	6.4	5.8	5.2	4.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	8	6.3	5.7	5.2	4.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.7	1.6	1.8	0.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.6	1.5	1.7	0.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.6	1.6	1.8	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.6	1.5	1.7	0.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.7	1.6	1.8	0.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.6	1.6	1.7	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.7	1.6	1.8	0.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.6	1.5	1.7	0.8	
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.1	39.8	14.9	5.2	1.4	20.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	19	42.2	13.4	4	0.6	20.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17	42.6	13.6	5.1	1.7	20
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.9	41.4	13.2	4.3	1	20.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.4	41.4	13.7	5.3	1.5	20.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.7	42.2	12.7	4.6	1	20.8
U2 - Efficacy additive alpha prior Beta(1, 1)	18.3	40.3	14.2	5.2	1.8	20.2
U3 - Efficacy additive alpha prior Beta(1, 1)	19.4	42.5	12.4	4	1.3	20.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.3	10.3	4.7	2	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.6	10.8	4.3	1.8	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.3	10.5	4.6	2	0.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.6	10.8	4.4	1.9	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.3	10.4	4.5	2	0.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.5	10.7	4.4	1.9	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.4	10.5	4.5	2	0.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.5	10.8	4.4	1.8	0.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.9	1.1	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.9	1	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	4	1	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4	1	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	4	1.1	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.9	1	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.9	1	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	4	1	0.5	0.1	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	65.4	23.7	4.7	0	0	6.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	69.5	20.9	3.4	0	0	6.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	65.9	23	4.7	0.2	0	6.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	70	19.9	3.5	0.1	0	6.5
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	65.9	23.4	4.5	0.1	0	6.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	68	21.6	4	0	0	6.4
U2 - Efficacy additive alpha prior Beta(1, 1)	65.8	23.3	4.8	0.2	0	5.9
U3 - Efficacy additive alpha prior Beta(1, 1)	69.2	20.9	3.8	0	0	6.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.5	8	3.2	1.2	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.2	7.7	2.9	1.1	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.6	7.9	3.2	1.2	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.2	7.6	2.9	1.2	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.5	8	3.2	1.2	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.1	7.6	3	1.2	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	16.6	7.9	3.3	1.2	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	17.1	7.7	3	1.2	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.2	3.5	0.9	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.2	3.4	0.8	0.4	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.2	3.4	0.8	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.2	3.4	0.8	0.4	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.2	3.4	0.9	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.2	3.4	0.8	0.4	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	1.2	3.4	0.9	0.4	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.2	3.4	0.8	0.4	0	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.8	14.8	11.7	40	19.4	0.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.9	14.9	10.4	37.7	16.8	0.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.5	14.7	12	40.1	19.4	0.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.3	14.6	10.7	37.9	17.2	0.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.8	14.1	11.2	41.5	19.1	0.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.3	15.7	10.4	37.3	17	0.3
U2 - Efficacy additive alpha prior Beta(1, 1)	13.6	14.9	11.5	40.1	19.6	0.3
U3 - Efficacy additive alpha prior Beta(1, 1)	20.1	15.3	10.5	36.8	17	0.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.7	5.4	5.7	9	5.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.1	5.5	5.4	9.2	4.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.6	5.3	5.7	9	5.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5	5.6	5.5	9.2	4.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.6	5.3	5.7	9.1	5.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5	5.5	5.5	9.1	4.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.6	5.4	5.7	9.1	5.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	5	5.5	5.5	9.1	4.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.1	1.6	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.2	1.6	2.3	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.1	1.6	2.4	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.2	1.6	2.3	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.1	1.6	2.4	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.2	1.6	2.3	1.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.1	1.6	2.4	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.1	1.6	2.3	1.1	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.1	40.5	13.8	6.3	1.6	17.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.9	42.5	11.8	5.1	1	16.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.3	42	13.6	6.4	1.2	17.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	23.2	42.3	11.4	4.9	1.1	17.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.7	40.9	13.9	6.3	1.4	16.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	22.3	42.3	12.2	5	1	17.2
U2 - Efficacy additive alpha prior Beta(1, 1)	20.9	40.6	13.8	6.3	1.3	17.1
U3 - Efficacy additive alpha prior Beta(1, 1)	22.2	42.6	12.1	4.8	1	17.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.2	9.7	4.8	2.8	0.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.6	10	4.6	2.5	0.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.2	9.7	4.8	2.8	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.6	9.9	4.6	2.6	0.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.1	9.9	4.7	2.8	0.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.5	9.9	4.7	2.6	0.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.2	9.8	4.9	2.8	0.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.5	9.9	4.6	2.6	0.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	4	1.2	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	4.1	1.1	0.8	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4.1	1.2	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4.1	1.1	0.8	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.1	1.2	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.1	1.2	0.8	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	4.1	1.2	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	4.1	1.2	0.8	0.2	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.8	18.6	15.9	28.8	12.9	1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	28.8	20.7	13.2	27.1	9.1	1.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	22.5	17.4	17.2	28.5	13.2	1.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	29.1	21	13.6	25.7	9.6	1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.2	18.8	17.2	28.7	13.2	0.9
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	30.3	20.3	13.3	25.8	9.4	0.9
U2 - Efficacy additive alpha prior Beta(1, 1)	21.7	19.4	17	27.9	13	1
U3 - Efficacy additive alpha prior Beta(1, 1)	29.4	21	12.9	26.7	9.1	0.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.2	6.3	5.9	7.3	4.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.8	6.4	5.6	7.2	3.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.3	6.4	5.8	7.2	4.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.8	6.4	5.6	7.3	3.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.2	6.4	5.8	7.3	4.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.7	6.4	5.6	7.2	4	
U2 - Efficacy additive alpha prior Beta(1, 1)	6.2	6.3	5.9	7.2	4.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	6.8	6.4	5.5	7.2	4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.6	1.7	2.2	0.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	3.6	1.7	2.1	0.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.6	1.7	2.2	0.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.6	1.6	2.1	0.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.6	1.7	2.2	0.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3.6	1.7	2.1	0.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.5	1.7	2.2	0.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	3.5	1.6	2.1	0.9	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	50.3	30.3	13.2	3.8	0.6	1.8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	55.3	29.6	9.9	2.6	0.5	2.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	49.8	30.6	13.1	4	0.5	2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	55.8	29.2	10.2	2.1	0.3	2.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	50.4	30.9	12.8	3.6	0.6	1.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	58.5	27.3	9.5	2.1	0.5	2.1
U2 - Efficacy additive alpha prior Beta(1, 1)	48.1	32.3	13.3	3.7	0.7	1.9
U3 - Efficacy additive alpha prior Beta(1, 1)	57.1	29.4	8.9	1.8	0.4	2.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.1	8.7	5.7	3.9	1.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11	8.4	5.3	3.8	1.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.1	8.6	5.8	3.9	1.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.1	8.3	5.3	3.8	1.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.1	8.7	5.8	3.9	1.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.1	8.4	5.3	3.7	1.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	10	8.7	5.8	3.9	1.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.1	8.4	5.3	3.7	1.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.2	4.2	1.8	1.4	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.3	4.1	1.7	1.4	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.2	4.2	1.8	1.4	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.3	4.1	1.7	1.4	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.2	4.2	1.8	1.4	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.3	4.1	1.7	1.4	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	1.1	4.2	1.8	1.4	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.4	4.1	1.7	1.4	0.4	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	38.2	25.3	16.7	11.8	8	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	43.6	28.7	14	9.4	4.3	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	37.9	26	16.9	11.9	7.3	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	44	27.8	14.6	8.6	5	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	38.1	25.8	16.3	12	7.8	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	44.4	27.4	14.9	8.8	4.5	0
U2 - Efficacy additive alpha prior Beta(1, 1)	37.8	26.5	16.6	11.6	7.5	0
U3 - Efficacy additive alpha prior Beta(1, 1)	44	27	15.2	9.8	4	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.3	5	6.1	8	6.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.6	5.5	6.3	8	5.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.4	5	6.1	8.1	6.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.7	5.5	6.2	8	5.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.3	5.1	6.1	8.2	6.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.6	5.5	6.3	7.9	5.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.3	4.9	6.1	8.2	6.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.7	5.4	6.2	8	5.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	2.2	2.9	1.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.9	2.3	2.9	1.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	2.1	3	1.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.9	2.3	2.9	1.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.8	2.1	3	1.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.9	2.3	2.9	1.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.8	2.1	3	1.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.9	2.2	2.9	1.4	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.1	59.8	16.8	1.2	0	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.8	62.9	13.9	1.3	0	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.6	60.9	17.4	1	0	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	22.7	63.5	12.6	1.1	0	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21	61.7	16.2	0.9	0	0.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	23.3	61.3	14.3	0.9	0	0.2
U2 - Efficacy additive alpha prior Beta(1, 1)	21.3	61	16.5	1	0.1	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	23.4	62.4	13.4	0.7	0	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.3	14.6	5.1	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.5	15.3	4.4	0.7	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.1	14.6	5.3	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.6	15.3	4.4	0.7	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.1	14.8	5.1	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.6	15.3	4.3	0.6	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.1	14.8	5.1	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.6	15.1	4.5	0.7	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.9	1.3	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.9	1.1	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.9	1.3	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.9	1.1	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.9	1.3	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.9	1.1	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.9	1.3	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.9	1.1	0.2	0	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.2	10.2	10.4	16.8	50.4	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.6	12.5	9.6	13.8	45.5	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.5	9.9	10.7	16.8	50.1	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.1	11.9	10.4	13.3	44.3	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.4	8.9	10.4	17.2	50.1	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.1	11.7	11	13.7	43.5	0
U2 - Efficacy additive alpha prior Beta(1, 1)	12.1	9.2	10.1	17.5	51.1	0
U3 - Efficacy additive alpha prior Beta(1, 1)	20.2	12.2	10.6	13.4	43.6	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.4	3.9	5.2	6.9	10.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.6	4.2	5.2	7	10.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.4	3.9	5.1	7	10.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.6	4.1	5.2	7	10.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.4	3.9	5.1	6.9	10.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.6	4.1	5.3	7	10	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.4	3.9	5.1	6.9	10.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.5	4.1	5.3	7.1	10	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.7	1.5	2.6	1.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	1.6	2.5	1.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.5	2.5	1.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	1.6	2.5	1.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.7	1.5	2.5	1.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.8	1.6	2.5	1.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.5	2.5	1.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.8	1.6	2.5	1.6	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.7	57.8	17.6	1.4	0.1	2.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	23.5	58.7	14.5	0.6	0.1	2.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.6	59.2	16.2	1.6	0	2.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	23	58.2	14.9	1.3	0	2.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.2	58.7	16	1.4	0.1	2.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	23.3	58	14.5	1.2	0.1	2.9
U2 - Efficacy additive alpha prior Beta(1, 1)	21.3	58.1	17	1.1	0.1	2.4
U3 - Efficacy additive alpha prior Beta(1, 1)	22.7	60	13.9	1.1	0.1	2.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.2	12.5	6.5	2.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.6	12.7	6.1	2.1	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.2	12.5	6.4	2.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.6	12.8	6.1	2.1	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.3	12.4	6.5	2.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.6	12.7	6	2.1	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.3	12.5	6.4	2.3	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	8.6	12.8	6.1	2.1	0.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	4.6	1.7	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	4.6	1.6	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4.6	1.7	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4.6	1.6	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.6	1.7	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.6	1.6	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	4.6	1.7	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	4.6	1.6	0.7	0.1	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.3	49.3	26.9	9.7	1.5	0.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.9	52.1	23.2	8.8	0.7	0.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.9	48.2	28.1	9.3	1.2	0.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15	50.6	24.4	8.8	0.8	0.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.8	46.5	29.3	10	1.1	0.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.4	52.5	23	8.8	1	0.3
U2 - Efficacy additive alpha prior Beta(1, 1)	12.9	47.3	27.6	10.5	1.4	0.3
U3 - Efficacy additive alpha prior Beta(1, 1)	15.5	51.5	24	7.8	0.8	0.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.9	10	8.2	5.1	1.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.3	10.5	7.8	4.8	1.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5	9.9	8.3	5	1.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.3	10.5	7.9	4.7	1.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.9	9.9	8.4	5.1	1.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.3	10.6	7.8	4.7	1.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.9	10	8.3	5.1	1.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.4	10.5	7.9	4.7	1.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.9	2.3	1.7	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.9	2.2	1.6	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.8	2.3	1.7	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.9	2.2	1.6	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.9	2.3	1.7	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.9	2.2	1.6	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.8	2.3	1.7	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.9	2.2	1.6	0.4	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	54.9	2.1	0	0	0	43
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	55	1.9	0	0	0	43.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	54.9	2.1	0	0	0	43
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	55	1.6	0	0	0	43.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	55	1.8	0	0	0	43.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	56.4	1.5	0	0	0	42.1
U2 - Efficacy additive alpha prior Beta(1, 1)	54.3	2.4	0	0	0	43.3
U3 - Efficacy additive alpha prior Beta(1, 1)	54.8	2	0	0	0	43.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.7	2.8	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.7	2.8	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.7	2.8	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.7	2.7	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.6	2.8	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.9	2.7	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	19.6	2.9	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	19.7	2.8	0.1	0	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	1.1	0	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	1.1	0	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	1.1	0	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	1.1	0	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	1.1	0	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	1	0	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	1.1	0	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	1.1	0	0	0	
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	23.5	59.9	12.7	2.2	0	1.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	27.7	55.6	13	2	0	1.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	23.1	59.1	13.9	2	0.1	1.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	29.7	54.6	12.1	1.8	0	1.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	23.9	58.7	13.8	1.8	0.1	1.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	29	54.9	12.4	1.9	0	1.8
U2 - Efficacy additive alpha prior Beta(1, 1)	23.4	58.9	13.6	2.3	0.1	1.7
U3 - Efficacy additive alpha prior Beta(1, 1)	28	55.7	12.6	1.8	0	1.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.9	12.8	5.8	2.3	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.3	12.7	5.6	2.1	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.9	12.7	5.8	2.2	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.3	12.7	5.5	2.1	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.9	12.8	5.8	2.2	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.4	12.7	5.5	2.1	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.9	12.7	5.8	2.3	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.4	12.7	5.5	2.1	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	4.7	1.6	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.2	4.6	1.6	0.6	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	4.6	1.7	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	4.6	1.6	0.6	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	4.6	1.7	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.2	4.6	1.6	0.6	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	1.1	4.6	1.7	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.1	4.6	1.6	0.6	0	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.9	4	0.1	0	0	75
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.5	3.8	0.1	0	0	75.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.6	3.4	0.1	0	0	74.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.8	4.4	0	0	0	75.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.6	4.3	0	0	0	75.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.1	4.3	0.1	0	0	75.5
U2 - Efficacy additive alpha prior Beta(1, 1)	20.7	3.7	0.1	0	0	75.5
U3 - Efficacy additive alpha prior Beta(1, 1)	21.1	4.1	0.1	0	0	74.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.2	5	1	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.2	4.8	1	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.2	4.9	1	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.1	4.8	1	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.1	4.9	1	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.2	4.8	1	0.1	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	15.1	5	1	0.1	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	15.2	4.8	1	0.1	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	2.3	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	2.3	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	2.3	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	2.3	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	2.3	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	2.2	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	2.3	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	2.2	0.2	0	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.6	0.9	3.9	49.6	44	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.4	1.4	4.1	52.5	38.6	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.6	0.9	3.9	50.1	43.5	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3	1	5.2	52.2	38.6	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.6	0.7	3.7	49.9	44.1	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	2.9	1.2	5.1	52.4	38.4	0
U2 - Efficacy additive alpha prior Beta(1, 1)	2	0.9	3.3	49.2	44.6	0
U3 - Efficacy additive alpha prior Beta(1, 1)	3.1	1.1	4.2	53.6	38	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.1	3.1	3.9	11.8	8.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.1	3.2	3.9	12.5	7.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.1	3.1	3.9	11.9	8.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.1	3.1	4	12.4	7.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.1	3.2	3.9	11.7	8.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.1	3.2	3.9	12.4	7.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.1	3.1	3.8	11.8	8.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.1	3.2	3.9	12.5	7.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0	2.4	1.1	2.5	1.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0	2.4	1.1	2.5	1.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0	2.4	1.1	2.4	1.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0	2.4	1.1	2.4	1.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0	2.4	1	2.4	1.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0	2.4	1.1	2.5	1.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	0	2.4	1.1	2.5	1.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	0	2.4	1.1	2.5	1.4	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	29.3	35	30.1	5.5	0	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	34.6	34.2	26.7	4.4	0	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	30.2	34.6	30.1	5	0	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	35.5	33.3	26.8	4.3	0	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	29.2	35.1	30.9	4.7	0	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	34	34.1	27.8	4	0	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	29.6	35.4	29.5	5.4	0	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	35.7	33.8	25.7	4.7	0	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.8	9.9	8.4	2.6	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.3	10	8.1	2.4	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.8	9.9	8.4	2.6	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.3	10	8	2.4	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.7	9.9	8.5	2.6	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.3	9.9	8.1	2.4	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.7	9.9	8.4	2.7	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.4	9.9	8.1	2.5	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	3.8	1.9	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	3.8	1.9	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	3.8	2	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	3.8	1.9	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3.9	2	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	3.9	1.9	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	3.9	1.9	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	3.8	1.9	0.9	0.1	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	27.1	35.5	29.1	5	0.1	3.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	33.4	34	25	4.4	0.2	3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.7	36.2	29	5.1	0.1	2.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	32.9	33.5	26.1	4.3	0.2	3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.9	35.6	29.4	4.8	0.1	3.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	31.9	33.5	26.2	4.9	0.2	3.3
U2 - Efficacy additive alpha prior Beta(1, 1)	26.2	37.1	28.9	4.8	0.1	2.9
U3 - Efficacy additive alpha prior Beta(1, 1)	32.4	34.7	25.8	4.1	0.2	2.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9	9.5	7.9	3	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.5	9.4	7.6	2.9	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9	9.4	7.9	3.1	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.6	9.4	7.6	2.8	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9	9.4	8	3	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.6	9.4	7.6	2.8	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.1	9.4	7.8	3.1	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.6	9.4	7.6	2.8	0.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	4.1	1.7	1.1	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	4	1.6	1.1	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4.1	1.7	1.1	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	4.1	1.6	1	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.1	1.7	1.1	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	4	1.6	1	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	4.1	1.7	1.1	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	4.1	1.6	1	0.1	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.1	30.5	39.6	9.5	0.5	0.8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	23.1	30.5	36.6	8.7	0.3	0.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18	31.1	39.7	9.8	0.5	0.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	22	30.4	38	8.5	0.2	0.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.6	32.1	40.3	8.8	0.4	0.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	23.5	30.4	37.4	7.5	0.4	0.8
U2 - Efficacy additive alpha prior Beta(1, 1)	17.1	30.8	41.6	9.1	0.5	0.9
U3 - Efficacy additive alpha prior Beta(1, 1)	21	32.4	37	8.3	0.3	1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.8	8.6	9.8	4	0.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.2	8.7	9.6	3.7	0.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.7	8.5	9.9	4	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.2	8.7	9.7	3.7	0.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.7	8.7	9.8	4	0.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.2	8.6	9.8	3.7	0.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	6.8	8.6	9.9	4	0.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	7.2	8.9	9.6	3.7	0.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.8	2.1	1.4	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.7	2	1.3	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.7	2.1	1.4	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.8	2.1	1.2	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.8	2.1	1.3	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.7	2	1.3	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.8	2.1	1.3	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.8	2.1	1.3	0.1	
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	29	46.9	5.1	1	0	18
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	30.2	45.8	5.4	0.3	0	18.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	27.2	47.1	6.6	0.7	0	18.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	29.5	46.8	4.7	0.2	0	18.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	27.3	46.5	7	0.6	0	18.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	29.6	45.3	5.6	0.3	0	19.2
U2 - Efficacy additive alpha prior Beta(1, 1)	28.4	46.8	6.1	0.3	0	18.4
U3 - Efficacy additive alpha prior Beta(1, 1)	28.7	46.3	5.1	0.2	0	19.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.8	11.4	3.9	1.1	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.2	11.5	3.6	1	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.8	11.4	3.9	1.1	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.1	11.5	3.6	1	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.8	11.4	3.8	1.1	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.1	11.3	3.7	1.1	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	11.9	11.4	3.9	1.1	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	12.2	11.4	3.6	1	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	4.4	1.1	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	4.4	1	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	4.4	1.1	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	4.3	1	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	4.4	1.1	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	4.3	1	0.3	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	1.1	4.4	1.1	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.1	4.3	1	0.3	0	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	60.9	31.9	4.8	0.1	0	2.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	63.5	30.5	3.6	0	0	2.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	62.1	31.2	4.3	0	0	2.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	63.4	30.9	3.3	0	0	2.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	60.9	31.5	5.1	0.1	0	2.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	65.2	28.9	3.5	0.1	0	2.3
U2 - Efficacy additive alpha prior Beta(1, 1)	61.1	31.9	4.6	0.1	0	2.3
U3 - Efficacy additive alpha prior Beta(1, 1)	64.7	29.9	3.1	0	0	2.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.4	9.4	2.3	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.1	9.2	1.9	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.4	9.4	2.3	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.1	9.3	1.9	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.2	9.5	2.4	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.1	9.2	1.9	0.3	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	17.3	9.4	2.4	0.4	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	17.9	9.4	1.9	0.3	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	3.2	0.8	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	3	0.7	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	3.2	0.8	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	3	0.7	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3.2	0.8	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	3	0.7	0.1	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.9	3.2	0.9	0.1	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	3.1	0.7	0.1	0	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.2	15.3	13.7	39.6	12.2	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	26.1	16.9	12.5	34.1	10.4	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.9	13.6	14.3	40.6	12.6	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	26	15.7	13.1	34.8	10.4	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.6	13.4	14	39.6	12.4	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	25.9	17.5	13.1	34.1	9.4	0
U2 - Efficacy additive alpha prior Beta(1, 1)	19.4	14.2	15.4	38.7	12.3	0
U3 - Efficacy additive alpha prior Beta(1, 1)	25.1	17.2	12.8	35.1	9.8	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.4	5	6.3	10.1	4.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.7	5.3	6.2	9.9	3.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.4	5	6.2	10	4.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.7	5.3	6.2	9.9	3.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.4	5	6.3	9.9	4.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.7	5.4	6.3	9.8	3.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.4	5	6.2	10	4.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.7	5.3	6.2	10	3.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.8	1.8	2.4	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.9	1.9	2.3	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.8	1.9	2.3	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.9	1.9	2.3	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.8	1.8	2.3	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.9	1.9	2.3	1.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.8	1.8	2.4	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.9	1.9	2.3	1.1	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.1	33.3	29	27.9	6.7	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5	38.9	29.5	22.9	3.7	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3	32.7	31.2	27.8	5.3	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.8	40.2	28.8	23.4	2.8	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.2	33.4	31.8	26.4	5.2	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.1	39	31.9	22.1	2.9	0
U2 - Efficacy additive alpha prior Beta(1, 1)	3.3	33.3	30.5	27.1	5.8	0
U3 - Efficacy additive alpha prior Beta(1, 1)	4.8	38.2	29.6	24.3	3.1	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.8	11.1	8.1	5.5	1.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4	12.2	8	4.8	1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.8	11	8	5.6	1.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.9	12.3	8	4.9	0.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.8	11.1	8.2	5.5	1.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.9	12.2	8.2	4.8	0.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.8	11.1	8.2	5.4	1.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.9	12	8.1	5	0.9	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.6	1.6	1.3	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.8	1.6	1.1	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.6	1.3	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.8	1.6	1.2	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0	2.6	1.7	1.3	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.8	1.7	1.1	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.7	1.3	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.8	1.7	1.2	0.3	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.4	13.7	31.8	28.6	12.5	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.6	18.1	31.4	24.7	7.2	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.2	12.8	32.4	28.6	13	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.6	17.2	31.8	24.9	6.5	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	13	13.2	31.9	29.5	12.4	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.2	17.8	31.9	24.8	6.3	0
U2 - Efficacy additive alpha prior Beta(1, 1)	13.6	13.8	31.6	28.5	12.5	0
U3 - Efficacy additive alpha prior Beta(1, 1)	18.9	18.2	31.3	25.2	6.4	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.5	4.2	8.3	8.9	5.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.7	4.6	8.5	8.8	4.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.5	4.2	8.3	8.8	5.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.6	4.6	8.5	8.8	4.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.5	4.2	8.3	8.8	5.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.6	4.6	8.6	8.7	4.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.5	4.2	8.3	8.8	5.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.6	4.6	8.5	8.8	4.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.7	1.9	2.9	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	2	2.8	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.9	2.9	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	2	2.8	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.7	1.9	3	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.8	2	2.8	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.9	2.9	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.8	2	2.8	1	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9	6.1	13.5	43	28.4	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.5	7.2	13	40.6	24.7	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10	5.1	13.6	43.1	28.2	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.9	7.8	12.8	42.2	23.3	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.8	5	14	42.1	29.1	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.6	7.7	12.8	41.4	23.5	0
U2 - Efficacy additive alpha prior Beta(1, 1)	9.4	6.1	12.4	42.8	29.3	0
U3 - Efficacy additive alpha prior Beta(1, 1)	14.8	7.6	13.1	41.2	23.3	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.4	3.7	5.9	10.2	6.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.5	3.9	5.9	10.5	6.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.3	3.7	6	10.1	6.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.5	3.8	5.9	10.6	6.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.3	3.7	5.9	10.1	7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.5	3.8	5.9	10.6	6.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.3	3.7	5.9	10.1	6.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.5	3.8	5.9	10.6	6.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.7	1.5	2.8	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.7	1.5	2.7	1.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.6	1.5	2.8	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.6	2.7	1.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.7	1.5	2.8	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.7	1.5	2.7	1.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.5	2.7	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.6	2.8	1.2	
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17	15.5	42.2	20.6	4.5	0.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.1	16.1	41.9	17.6	3.1	0.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.3	15.4	43.5	19.9	4.7	0.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.4	17.5	40.8	17.6	3.5	0.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17	14.6	44	19.3	4.9	0.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	21	18	39.8	17.7	3.3	0.2
U2 - Efficacy additive alpha prior Beta(1, 1)	17.2	14.5	44.3	19.8	4	0.2
U3 - Efficacy additive alpha prior Beta(1, 1)	22.4	16.4	39.2	18.2	3.6	0.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.3	6.7	9.2	6.3	2.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.9	6.7	9.1	6.2	2.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.2	6.6	9.3	6.4	2.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.9	6.6	9.1	6.2	2.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.3	6.6	9.2	6.3	2.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.9	6.7	9.1	6.2	2.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.3	6.6	9.3	6.4	2.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.8	6.7	9	6.2	2.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.5	2.1	2	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.5	2.1	1.9	0.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.5	2.2	2	0.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.5	2.1	1.9	0.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.5	2.1	2	0.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.5	2.1	1.9	0.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.5	2.1	2	0.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.5	2.1	1.9	0.6	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.6	39.2	34.5	13.6	3.1	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.2	45.5	32	9.8	1.5	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8	41.1	34.7	13.4	2.8	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11	43.2	31.7	12	2.1	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.4	39.2	35.1	14.6	2.7	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.8	44	32.1	10.7	1.4	0
U2 - Efficacy additive alpha prior Beta(1, 1)	9.1	40	34.2	13.9	2.8	0
U3 - Efficacy additive alpha prior Beta(1, 1)	12.3	43.7	31.6	10.7	1.7	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.4	11.3	9	4.2	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.6	12.6	8.6	3.4	0.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.5	11.3	9	4	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.6	12.3	8.7	3.5	0.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.4	11.3	9	4.1	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.6	12.4	8.7	3.5	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.5	11.4	8.9	4	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.7	12.2	8.8	3.5	0.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.1	2.1	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.2	2	1	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.1	2	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.2	2	1	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3	2.1	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.1	2	1.1	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.1	2	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.2	2	1	0.2	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	24.6	44.4	15.7	6.3	2.7	6.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	29.6	43.3	14.1	5.2	1.4	6.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.2	44.2	14.8	5.9	2.9	6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	28.4	44.9	13	5.8	1.7	6.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.3	44.6	15.1	5.7	2.6	5.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	29.4	43.8	12.9	5.5	2.1	6.3
U2 - Efficacy additive alpha prior Beta(1, 1)	27.7	43.9	14.8	6	2	5.6
U3 - Efficacy additive alpha prior Beta(1, 1)	30.2	43.6	13.2	5.4	1.5	6.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9	10.2	5.5	3.5	1.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.4	10.2	5.4	3.4	1.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9	10.2	5.6	3.4	1.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.4	10.2	5.3	3.4	1.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.9	10.3	5.6	3.4	1.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.5	10.1	5.3	3.4	1.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.9	10.3	5.5	3.4	1.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.4	10.1	5.4	3.4	1.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	4.5	1.5	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	4.4	1.5	1.1	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	4.5	1.5	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	4.5	1.5	1.1	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	4.5	1.5	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	4.5	1.5	1	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.9	4.6	1.4	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	1	4.5	1.5	1.1	0.3	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	25.8	52.1	15.5	3.6	0.3	2.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	30.4	52	11.9	2.7	0.1	2.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	25.8	53.1	14.8	3.3	0.4	2.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	29.5	52.7	12.3	2.6	0.3	2.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.4	52	15.2	3.2	0.5	2.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	29.8	52.4	12.3	2.4	0.2	2.9
U2 - Efficacy additive alpha prior Beta(1, 1)	26.4	52.7	14.8	3.1	0.5	2.5
U3 - Efficacy additive alpha prior Beta(1, 1)	29.3	53	12.4	2.5	0.2	2.6
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.4	11.4	5.9	3.1	1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.9	11.3	5.6	3	0.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.5	11.3	6	3.1	1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9	11.3	5.6	3	0.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.4	11.3	5.9	3.2	1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.1	11.3	5.6	3	0.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.4	11.3	6	3.1	1	
U3 - Efficacy additive alpha prior Beta(1, 1)	8.9	11.4	5.6	3	0.9	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	4.8	1.6	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	4.8	1.5	1	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	4.8	1.7	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	4.8	1.5	1	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	4.8	1.6	1.1	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	4.8	1.6	1	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	1	4.8	1.6	1.1	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.1	4.7	1.5	1	0.2	